Modelling and Forecasting the Underwriting Cycle in General Insurance Using Hidden Markov Switching Time Series Models

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Submitted as the Final Year Project for the BSc Honours Degree in Actuarial Science, City St George's, University of London.

16 April 2026



DECLARATION

I certify that I have complied with the guidelines on plagiarism outlined in the Course Handbook in the production of this dissertation and that it is my own, unaided work.

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16 April 2026

ACKNOWLEDGEMENTS

I am sincerely grateful to my supervisor, Dr Iqbal Owadally, whose guidance, intellectual generosity, and patience shaped every stage of this project. His expertise in actuarial modelling and time series analysis provided both the methodological foundation and the data that made this research possible. I am especially thankful for his willingness to engage with my questions throughout the year and for his careful and constructive feedback on successive drafts.

I would like to thank the Faculty of Actuarial Science and Insurance at Bayes Business School for the academic environment and resources that supported this work.

Finally, my deepest thanks go to my family and friends. Their encouragement, understanding, and unwavering support throughout my studies have meant more than I can properly express here.

ABSTRACT

Insurance markets do not drift; they switch. This research investigates the underwriting cycle in the Canadian Property and Casualty (P&C) market over 1985-2022 using Hidden Markov Switching Models (HMSMs) on Personal Property (PP) and Auto loss ratios. By modelling loss ratios as driven by latent states with regime-dependent means and variances, HMMs allow underwriting regimes to be identified probabilistically and interpreted as hard and soft market conditions. Spectral and wavelet analyses show that PP follows irregular, time-varying cycles, whereas Auto displays a clearer decadal pattern. The preferred two-state HMM recovers economically intuitive regimes: long, hard-market phases of roughly 10 years for PP and extended, benign periods of about 7 years for Auto, consistent with the frequency-domain evidence. Overall, regime-switching models provide a more informative and practical description of underwriting cycle dynamics than linear benchmarks, supporting cycle-aware pricing, reserving, and capital planning in P&C insurance.

Contents

1. INTRODUCTION 1
2. LITERATURE REVIEW 3
   2.1 Classical Empirical Foundations 3
   2.2 Economic Drivers and Theoretical Frameworks 3
   2.3 Challenges to Linear Cycle Detection 3
   2.4 Regime-switching, Spectral, and Wavelet Methods 4
3. DATA AND METHODOLOGY 5
   3.1 Data Description 5
   3.2 Stationarity Testing 5
   3.3 ARIMA and GARCH Benchmark Models 5
   3.4 Spectral and Wavelet Analysis 6
   3.5 Hidden Markov Model (HMM) 6
4. DATA ANALYSIS AND RESULTS 7
   4.1 Descriptive Analysis 7
   4.2 Stationarity and Preliminary Tests 8
   4.3 Linear Mean Models (ARIMA) 9
   4.4 Volatility Models (GARCH) 12
   4.5 Cycle analysis (Spectrum and Wavelets) 13
      4.5.1 Personal Property 13
      4.5.2 Auto 14
   4.6 Hidden Markov Models (HMM) 16
      4.6.1 Personal Property 16
      4.6.2 Auto 17
      4.6.3 Regime Conditioned Forecasts and Practical Implications 19
5. CONCLUSION 22

REFERENCES 23

APPENDIX A – DATA SOURCE 26

APPENDIX B – R CODE 27

List of Figures

Figure 1: PP Loss Ratio, 1985–2022 7
Figure 2: Auto Loss Ratio, 1985-2022 8
Figure 3: Residual diagnostics for AR(1) model of PP loss ratio 10
Figure 4: Residual diagnostics for MA(1) model of ΔPP loss ratio 11
Figure 5: Residual diagnostics for ARIMA(0,1,1) for ΔAuto 11
Figure 6: Smoothed spectrum – PP loss ratio 13
Figure 7: Wavelet power spectrum – PP loss ratio 14
Figure 8: Smoothed spectrum – Auto loss ratio 15
Figure 9: Wavelet power spectrum – Auto loss ratio 15
Figure 10: Hidden Markov state classification of PP loss ratio over time 17
Figure 11: Auto GARCH residuals – HMM regime classification 18
Figure 12: Regime-conditioned forecasts 2023–2025 - PP 20
Figure 13: Regime-conditioned forecasts 2023–2025 – Auto 20

List of Tables

Table 1: Descriptive statistics - Canadian P&C loss ratios, 1985–2022 7
Table 2: Stationarity test results - ADF and KPSS 9
Table 3: ARIMA model selection - PP, ΔPP, ΔAuto 10
Table 4: GARCH(1,1) parameter estimates 12
Table 5: HMM parameter estimates - PP 16
Table 6: HMM parameter estimates – Auto 18
Table 7: Expected regime durations and stationary distributions 19

1. INTRODUCTION

Insurance plays a fundamental role in economic stability. It absorbs and redistributes risk across households, businesses, and governments. The underwriting cycle complicates this stabilising function in the Property and Casualty (P&C) industry. This cycle is a well-documented pattern of alternating hard and soft market phases, marked by variations in premium rates, loss ratios, and coverage availability (Harrington, 2004). Hard markets have elevated premiums, tighter underwriting standards, and reduced capacity. In contrast, soft markets feature competitive pricing, loosened standards, and surplus capacity. The ability to anticipate transitions between these phases affects pricing accuracy, reserve adequacy, capital allocation, and strategic planning.

Despite being recognised for over four decades, the underwriting cycle remains poorly understood. Its drivers are contested. Some cite capacity-constraint shocks following catastrophe losses (Harrington, 2004) or interest rate effects on investment income (Grace and Hotchkiss, 1995). Others focus on behavioural pricing errors in long-tailed lines and line-of-business heterogeneity in cycle length (Fung et al., 1998). More fundamentally, some dispute its very existence as a statistically stable phenomenon. Boyer et al. (2012) find that apparent cycles frequently fail to deliver meaningful out-of-sample forecast improvements. Boyer and Owadally (2015) argue that many detected cycles may be artefacts of model misspecification rather than genuine economic regularities.

These challenges expose a deeper inadequacy in the tools traditionally applied to this problem. AutoRegressive Integrated Moving Average (ARIMA), a time series forecasting model, and Generalised AutoRegressive Conditional Heteroskedasticity (GARCH), which models changes in market volatility, assume parameter stability across the full sample. This makes them structurally incapable of capturing the abrupt transitions between hard and soft market conditions that define the underwriting cycle (Hamilton, 1989). An insurer relying on a linear forecast during a regime transition will systematically misprice risk at precisely the moment accurate forecasting matters most.

This project fills the gap by applying Hidden Markov Switching Models (HMSMs), hereafter referred to as HMMs, to the Canadian P&C underwriting cycle from 1985-2022 and analysing Personal Property (PP) and Auto lines loss ratios separately. The central research question is whether Canadian P&C loss ratios are better described by persistent latent regimes or by stable linear dynamics, leading to two testable null hypotheses. The first posits that a single stable parameter set across the sample suffices, implying no significant regime switching beyond what linear ARIMA or GARCH models capture. The second tests whether any regime structure identified by the HMM improves out-of-sample forecast accuracy relative to linear models. Spectral and wavelet diagnostics then independently characterise cyclical structure for both lines,

thus providing cross-validation for the regime-switching findings (Torrence and Compo, 1998).

2. LITERATURE REVIEW

2.1 Classical Empirical Foundations

The underwriting cycle has been empirically studied for over four decades. Cummins and Outreville (1987) identified cycles in insurance profitability averaging 6-8 years across several OECD markets using autoregressive models, thereby establishing the notion of a persistent, quasi-regular feature of property-liability markets. Venezian (2006) extended this using spectral methods, detecting low-frequency cyclical components broadly consistent with that pattern. These early studies assumed the cycle was stable and symmetric, assumptions this project directly challenges.

2.2 Economic Drivers and Theoretical Frameworks

The capacity-constraint hypothesis holds that capital shocks create hard markets when underwriting capacity declines. Soft markets follow as capital rebuilds (Harrington, 2004; Doherty and Garven, 1995). Macroeconomic factors also play a role. Interest rates and GDP growth correlate with underwriting profitability, but provide limited short-term predictive power (Grace and Hotchkiss, 1995).

Cycle dynamics also vary by line-of-business. PP is strongly affected by catastrophe losses. These cause sharp but temporary increases in loss ratios. Auto profitability is more influenced by regulatory changes and claims inflation. These factors lead to more gradual performance shifts (Fung et al., 1998). Using vector autoregressive models, Fung et al. (1998) show that cycle length and amplitude differ across lines. This supports the need to analyse each line-of-business separately.

2.3 Challenges to Linear Cycle Detection

More recent work questions whether cycles are statistically identifiable at all. Boyer et al. (2012) show that business-cycle filtering approaches deliver little improvement in forecasting over simple AR benchmarks. Boyer and Owadally (2015) demonstrate that cyclical patterns detected in linear models can arise in random stationary processes, establishing a high evidential bar that any genuine cycle identification framework must clear. Born and Viscusi (2006) also question the practical predictability of underwriting cycles, highlighting the role of catastrophic shocks and insurer capacity adjustments in producing irregular, difficult-to-forecast dynamics.

2.4 Regime-switching, Spectral, and Wavelet Methods

In response, researchers have increasingly favoured regime-switching frameworks. For instance, Hamilton (1989) and Hardy (2001) demonstrated that economic and financial time series are better characterised by abrupt regime changes rather than stable linear dynamics; these findings are directly applicable to insurance markets, where conditions can shift rapidly following catastrophe events or regulatory changes. Similarly, Jawadi et al. (2009) apply non-linear cointegration methods to loss ratio modelling, finding that regime-based specifications capture dynamics that linear models miss entirely. Related regime-switching applications to insurance volatility are provided by Zhu and Liu (2012). Moreover, spectral analysis provides nonparametric evidence on cycle length (Venezian, 2006), while wavelet analysis addresses limitations of classical spectral methods by providing time-frequency decompositions that reveal how cyclical components evolve across the sample period (Torrence and Compo, 1998 ; Grinsted et al., 2004). Finally, Owadally et al. (2018, 2019) apply advanced signal processing techniques to insurance loss series, finding complex non-linear dynamics inconsistent with simple linear models, building on earlier work on insurance market dynamics (Owadally and Haberman, 2011).

Prior research shows that underwriting cycles differ across lines, are weakly detected by linear models, and are better captured by regime-switching frameworks supported by frequency-domain diagnostics. No study applies this approach to Canadian P&C data at the line-of-business level, quantifies regime durations and their consistency with spectral evidence, or translates HMM outputs into actuarially useful tools. This study fills that gap.

3. DATA AND METHODOLOGY

This section outlines the data and methodology for analysing and forecasting the underwriting cycle in the P&C insurance market. The analysis uses classical linear time-series models and non-linear HMMs. Exploratory diagnostics are applied in both the time and frequency domains. All statistical analysis is done in R.

3.1 Data Description

The primary dataset is annual aggregate loss ratios for PP and Auto lines in the Canadian P&C market. The sample spans 1985 to 2022 (n = 38). The loss ratio is net incurred claims divided by net written premiums. This is the standard actuarial metric for underwriting profitability and the main variable in cycle literature (Harrington, 2004). Data comes from several editions of the Insurance Bureau of Canada (IBC) Facts Book. Two lines are analysed separately because cycle dynamics vary widely across segments (Fung et al., 1998).

3.2 Stationarity Testing

Integration order relies on complementary tests with opposing null hypotheses. The Augmented Dickey-Fuller (ADF) test (Dickey and Fuller, 1979) tests for a unit root against stationarity. In contrast, the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test (Kwiatkowski et al., 1992) tests level stationarity against a unit root. Conflicting outcomes suggest near-integration, not contradiction; this is the situation for PP in levels, motivating analysis at both levels and first differences.

3.3 ARIMA and GARCH Benchmark Models

ARIMA models serve as the primary linear benchmarks. Orders are selected by the Akaike Information Criterion (AIC) (Akaike, 1974). Adequacy is evaluated using the Ljung-Box test (Box and Jenkins, 1976; Ljung and Box, 1978). When residuals show conditional heteroskedasticity, a GARCH(1,1) variance equation is added in the form:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

The condition $\alpha + \beta < 1$ ensures covariance stationarity. Values near 1 indicate persistent, nearly integrated volatility (Engle, 1982; Bollerslev, 1986; Tsay, 2010). These models serve as baselines for gauging the added value of regime-switching models.

3.4 Spectral and Wavelet Analysis

The smoothed power spectral density is estimated using the Daniell kernel (Priestley, 1981). The dominant frequency implies a cycle of period $\frac{1}{f}$ years (Priestley, 1981). A continuous Morlet wavelet transform is applied to obtain a time-frequency decomposition. This reveals how cyclical components evolve across the sample. Statistical significance is assessed against a red-noise background using 200 Monte Carlo simulations (Torrence and Compo, 1998; Percival and Walden, 2000).

3.5 Hidden Markov Model (HMM)

The HMM assumes the observed loss ratio follows:

$$Y_t \mid S_t = k \sim N(\mu_k, \sigma_k^2)$$

where St is a latent state governed by a first-order Markov chain with transition matrix, where the parameters are estimated by maximum likelihood via the Baum-Welch EM algorithm (Baum et al., 1970; Hamilton, 1989). Expected regime duration and the long-run stationary distribution follow directly:

$$E[T_k] = \frac{1}{1 - P_{kk}}$$

$$\pi_1 = \frac{(1 - P_{22})}{(2 - P_{11} - P_{22})}$$

Model selection between two-state and three-state specifications uses the Bayesian Information Criterion, $BIC = -2\ell(\hat{\theta}) + p \cdot \ln(n)$, where $p = K^2 + K$ free parameters (Schwarz, 1978). HMMs are estimated at levels for PP, where the near-stationary level series carries the economically meaningful regime structure, and at ARIMA-GARCH standardised residuals for Auto, where non-stationarity requires prior differencing before regime identification.

4. DATA ANALYSIS AND RESULTS

This section presents the empirical results for PP and Auto loss ratios from 1985 to 2022, with each stage contributing evidence toward the null hypotheses stated in Section 1.

4.1 Descriptive Analysis

The analysis begins with a descriptive examination of the PP and Auto loss ratio series. Summary statistics provide an initial comparison of central tendency and dispersion between the two lines of business.

Table 1: Descriptive statistics - Canadian P&C loss ratios, 1985–2022

SERIES	MIN	Q1	MEDIAN	MEAN	Q3	MAX	SD
PP	0.4899	0.5791	0.6321	0.6282	0.6663	0.7705	0.0649
AUTO	0.5777	0.7183	0.7497	0.7557	0.7915	0.9107	0.0732

Figure 1: PP Loss Ratio, 1985–2022

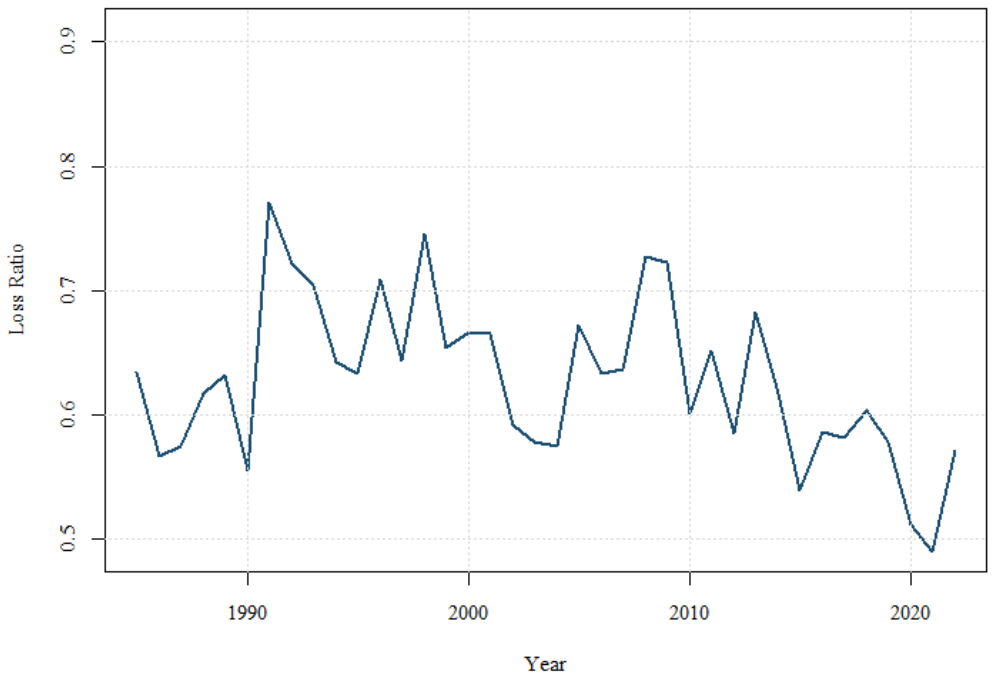
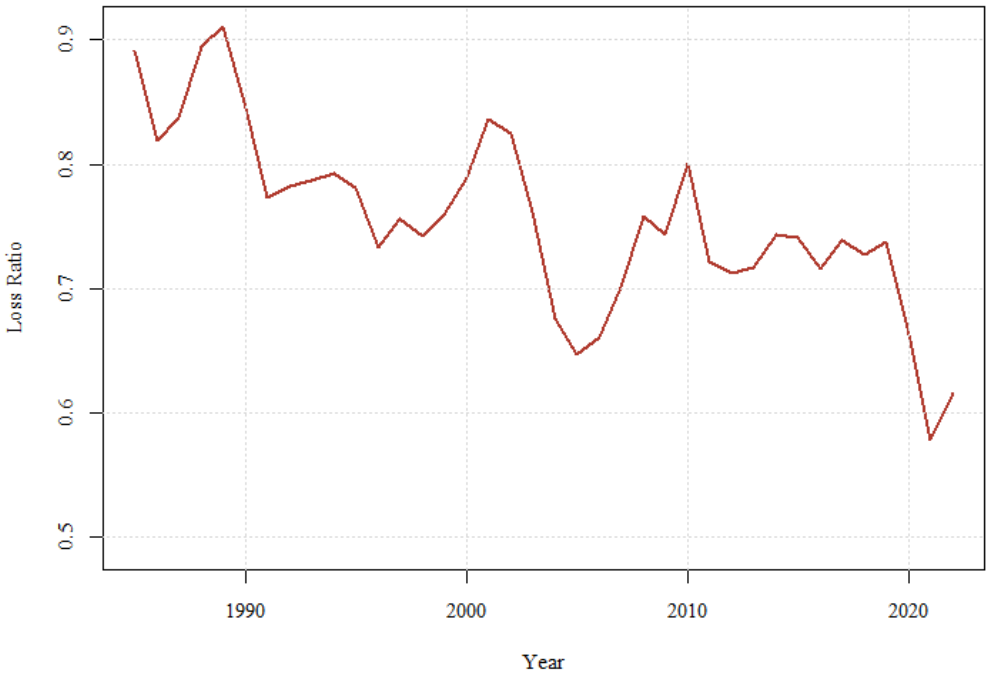


Figure 2: Auto Loss Ratio, 1985-2022



The descriptive statistics in Table 1 reveal a compelling contrast between the two lines. While PP is more profitable than Auto, its distinct statistical character is equally notable. Its values range from 0.490 to 0.771, a span of 0.281 points, or 45% of its mean, suggesting extended intervals of differing profitability rather than mere random noise. This persistent variation aligns with regime-switching processes (Hamilton, 1989). By comparison, Auto shows a higher mean of 0.756 and greater variability, highlighting its vulnerability to sustained reserve development and regulatory shocks rather than isolated single-year effects (Fung et al., 1998; Harrington, 2004). Both lines experience multi-year swings, deviating from simple white noise and providing initial descriptive evidence to challenge the first null hypothesis.

4.2 Stationarity and Preliminary Tests

This section evaluates the stationarity properties of each loss ratio series using complementary unit root tests. The results guide subsequent transformations used in ARIMA, GARCH, and HMM estimation.

Table 2: Stationarity test results - ADF and KPSS

SERIES	ADF STATISTIC	ADF P-VALUE	KPSS STATISTIC	KPSS P-VALUE	DECISION
PP	−2.621	0.331	0.392	0.081	Near-stationary
ΔPP	-4.070	< 0.001***	0.070	> 0.100	Stationary ✓
AUTO	−3.200	0.105	0.777	0.010	Non-stationary (I(1))
ΔAUTO	-4.210	0.013*	0.057	> 0.100	Stationary ✓

Note: ADF null hypothesis = unit root (non-stationary). KPSS null hypothesis = level stationarity. * p < 0.05; *** p < 0.001.

The stationarity results in Table 2 reveal distinct dynamics in the two lines. While PP exhibits a near-integrated profile, ADF fails to reject the unit root hypothesis, and KPSS fails to reject the null of stationarity. This combination indicates a series that is formally stationary but with autocorrelation decaying slowly enough to approximate a unit root over typical sample horizons (Harrington and Yu, 2003; Nelson and Plosser, 1982). Retaining PP at levels for HMM estimation is analytically justified: differencing would remove the low-frequency variation that encodes regime transitions. This low-frequency component is precisely where the predictive signal for shifts between market conditions resides (Hamilton, 1989).

Auto, by contrast, is unambiguously I(1), implying that shocks to its loss ratio permanently alter the trajectory of future outcomes, consistent with a structurally driven downward trend rather than cyclical fluctuations and explaining the failure of mean-reverting cycle detection methods (Boyer et al., 2012; Boyer and Owadally, 2015).

4.3 Linear Mean Models (ARIMA)

ARIMA models capture linear dependence in loss-ratio dynamics, providing a benchmark for persistence and short-run adjustment. Model adequacy is evaluated using standard diagnostics and information criteria.

Table 3: ARIMA model selection - PP, ΔPP, ΔAuto

SERIES	MODEL	AIC	COEFFICIENT	LJUNG-BOX P	RMSE	SELECTED
PP	AR(1)	-102.66	0.4237	0.9128	0.0578	✓
	MA (1)	-100.55			0.0595	
	ARMA(1,1)	-101.66			0.0570	
ΔPP	AR(1)	-93.938			0.0625	
	MA(1)	-97.901	-0.6455	0.9286	0.0590	✓
	ARMA(1,1)	-96.769			0.0565	
ΔAUTO	ARIMA(1,1,0)	-124.58			0.0420	
	ARIMA(0,1,1)	-124.95	0.2181	0.3297	0.0418	✓
	ARIMA(1,1,1)	-122.98			0.0418	

Figure 3: Residual diagnostics for AR(1) model of PP loss ratio

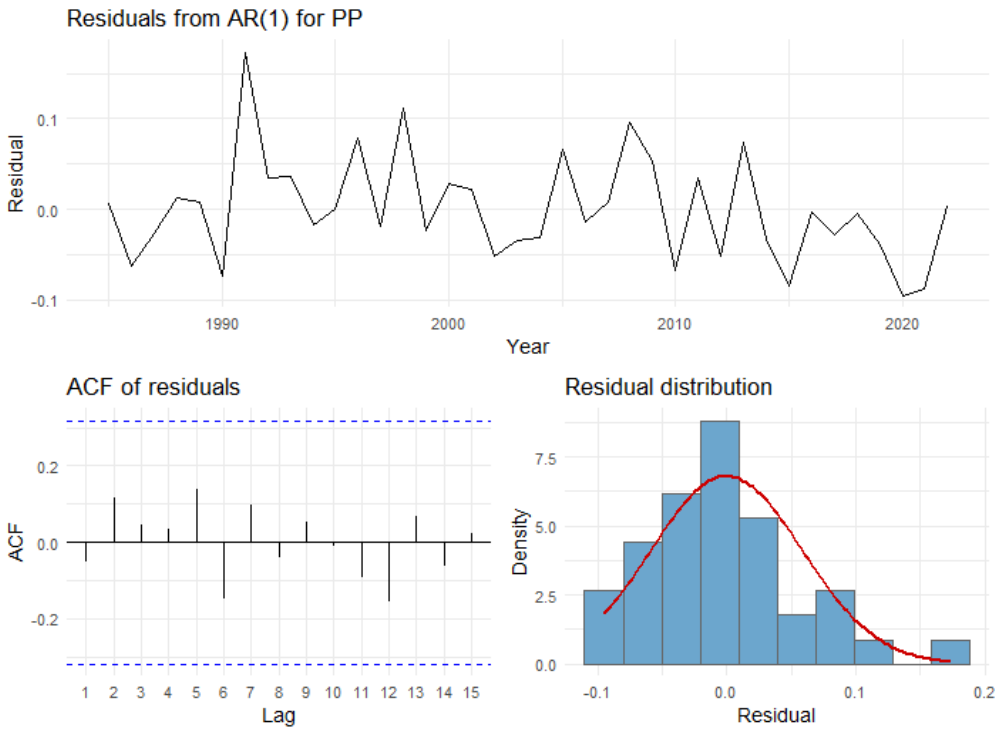


Figure 4: Residual diagnostics for MA(1) model of ΔPP loss ratio

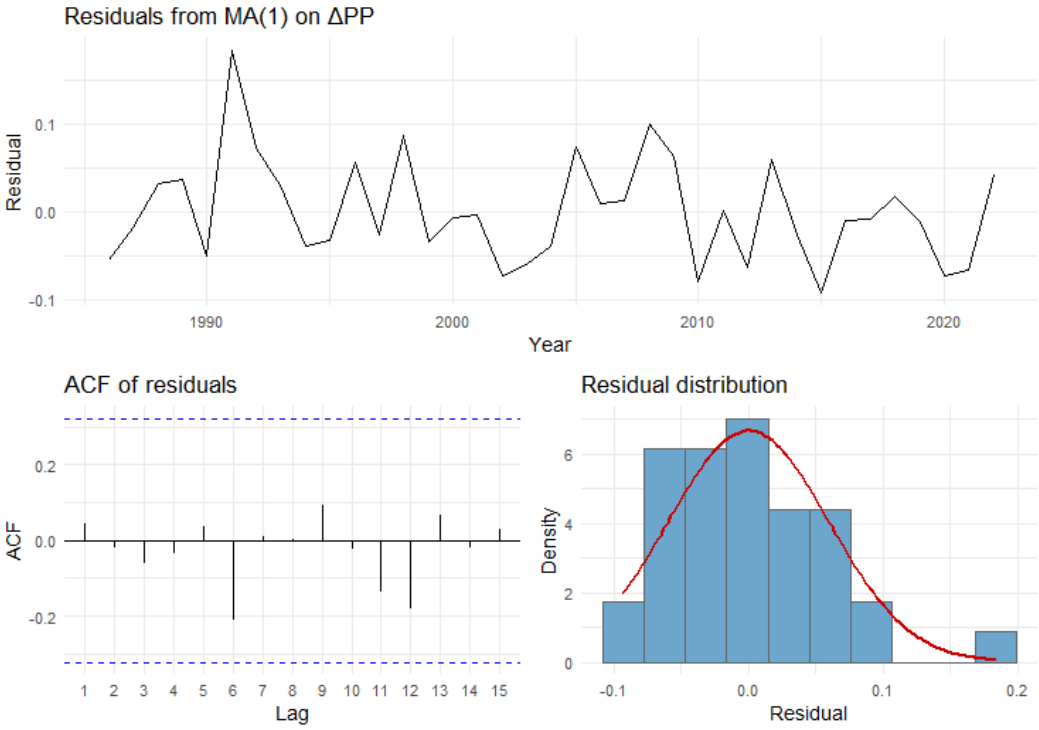
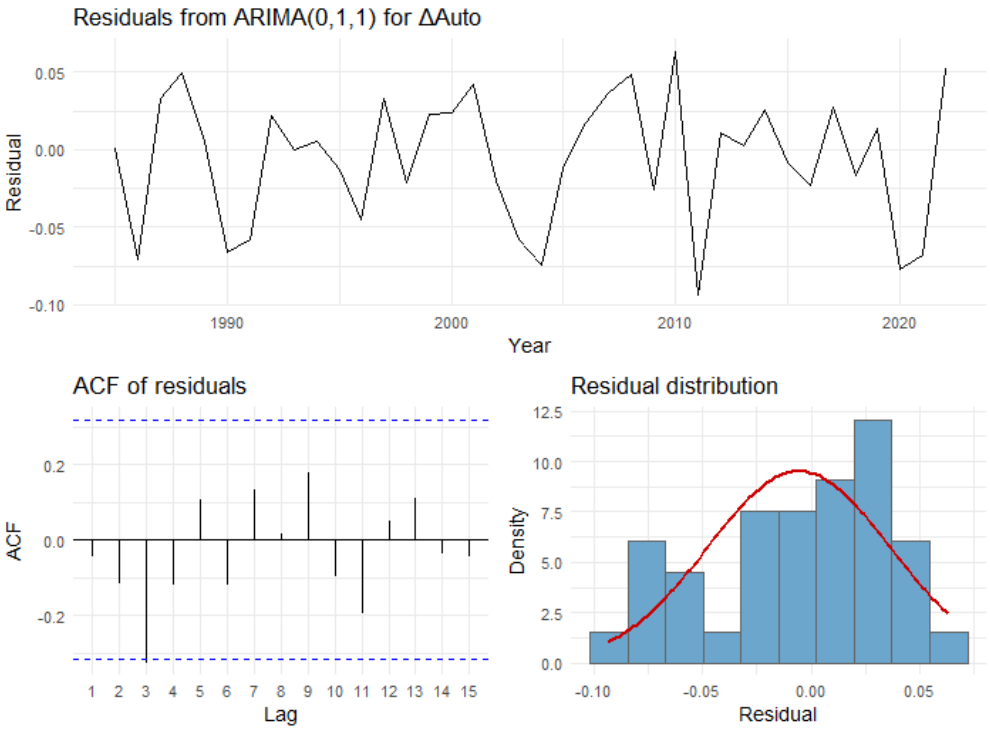


Figure 5: Residual diagnostics for ARIMA(0,1,1) for ΔAuto



The ARIMA results in Table 3 illustrate the inherent limits of linear models for capturing the structural dynamics of insurance loss ratios. While residuals are uncorrelated and the Ljung-Box test is non-significant, this apparent sufficiency obscures critical shortcomings. The coefficient of 0.424 implies symmetric, time-invariant persistence: each year's deviation from the mean carries forward identically, regardless of whether the market is in a prolonged hard phase or an emergent soft phase. However, such uniformity fails to capture the qualitative, regime-dependent dynamics central to actuarial decision-making (Harrington, 2004).

The MA(1) on ΔPP captures mild overshooting consistent with capacity-constrained price adjustments but treats this correction as unconditional and regime-invariant (Jawadi et al., 2009). The near-identical Root Mean Square Error (RMSE) for levels and first differences further suggests that differencing adds little value.

For Auto, ARIMA(0,1,1) exhausts the linear predictive content. Nearly identical AICs across alternative specifications indicate that the differenced series carries little linear structure beyond a single-period adjustment. This flat AIC surface itself is diagnostic, signalling that linear models cannot detect the non-linear, state-dependent regime dynamics that the HMM in Section 4.6 explicitly targets, reinforcing the need for models capable of representing regime shifts and asymmetric persistence (Hyndman and Athanasopoulos, 2021; Box and Jenkins, 1976; Brockwell and Davis, 2002).

4.4 Volatility Models (GARCH)

GARCH models are introduced to examine whether volatility varies systematically over time. This framework enables formal testing of persistence in conditional variance. The resulting residuals also provide inputs for later regime-switching analysis.

Table 4: GARCH(1,1) parameter estimates

SERIES	MODEL	α	β	α + β	RMSE	VS ARIMA
PP	AR(1)–GARCH(1,1)	≈ 0	0.9979	0.9979	0.0578	Similar
ΔPP	MA(1)–GARCH(1,1)	≈ 0	0.9933	0.9933	0.0591	Similar
ΔAUTO	ARIMA(0,1,1)–GARCH(1,1)	≈ 0	0.9990	0.9990	0.0419	Similar

The GARCH results in Table 4 highlight a clear structural distinction between the two lines. For PP, alpha is effectively zero in both specifications, so the model reduces to near constant variance. Rather than being a shortcoming, this finding confirms that annual property loss ratios lack volatility clustering typical of financial returns. Instead, temporal variation in variance is better explained by discrete regime shifts than by continuously evolving conditional volatility, thus justifying the HMM over GARCH for PP (Engle, 1982; Bollerslev, 1986; Tsay, 2010). For Auto, near-integrated volatility (with alpha + beta = 0.999) shows that shocks persist almost indefinitely, which aligns with the cumulative reserve uncertainty of lines with extended liability components. Although GARCH does not improve point forecast RMSE for either line, it does provide actuarially relevant volatility forecasts for Auto capital modelling, and its standardised residuals form the input to the Auto HMM in Section 4.6 (Bollerslev, 1986; Engle, 1982).

4.5 Cycle analysis (Spectrum and Wavelets)

4.5.1 Personal Property

Frequency-domain analysis of PP is conducted using spectral estimation to identify dominant periodic components and wavelet decomposition to capture time-varying cyclical behaviour.

Figure 6: Smoothed spectrum – PP loss ratio

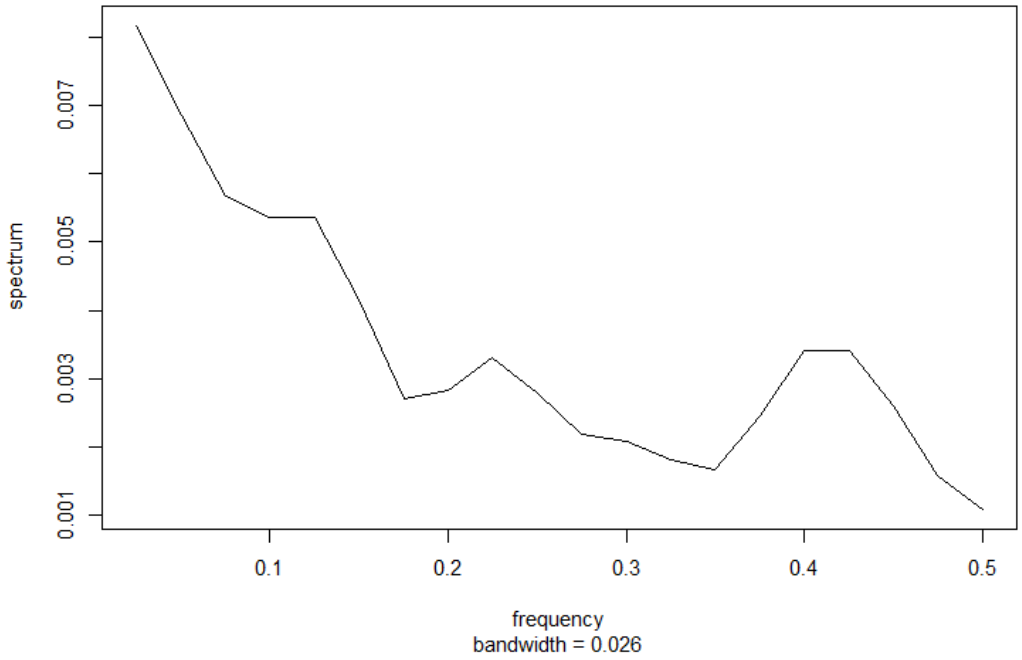
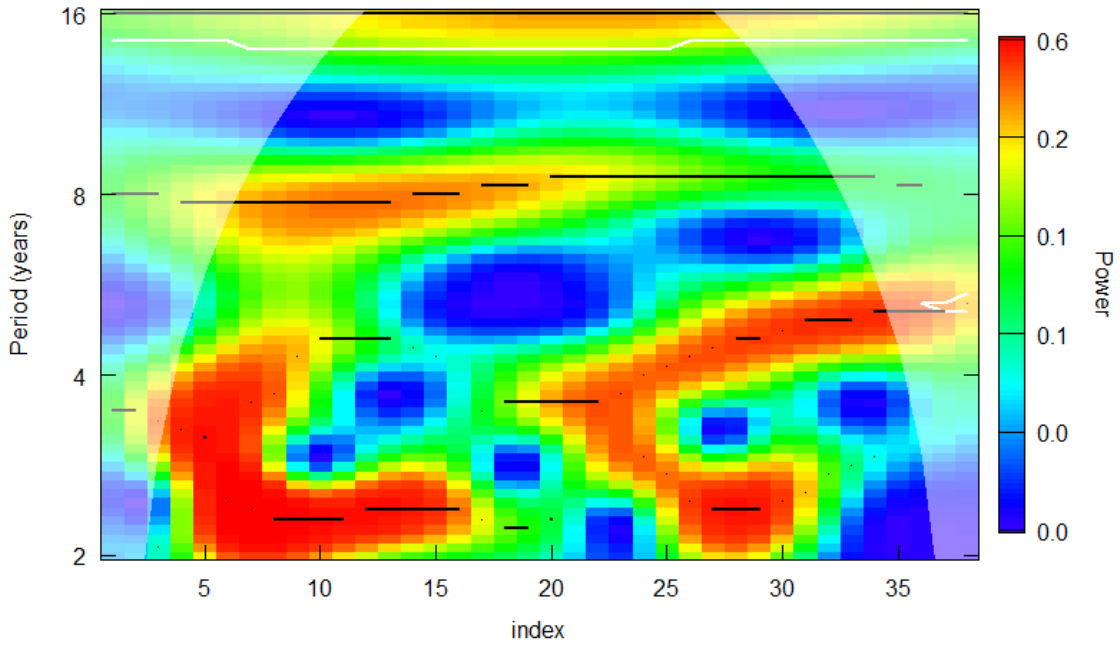


Figure 7: Wavelet power spectrum – PP loss ratio



The dominant spectral frequency of 0.025, corresponding to a 40-year period, is analytically significant not as evidence of a long cycle but precisely because it confirms the absence of any conventional underwriting cycle. No 6-to-8-year periodicity, as hypothesised by Cummins and Outreville (1987) and Venezian (2006) emerges, directly challenging classical fixed frequency characterisations. Wavelet decomposition reveals intermittent rather than persistent power at 2-to-6-year periods, reflecting a series that alternates between extended hard and soft market regimes and generating transient low-frequency power without a coherent spectral peak, a signature consistent with regime-switching dynamics (Torrence and Compo, 1998; Owadally et al., 2018).

4.5.2 Auto

The same frequency-domain techniques are applied to the Auto loss ratio series to allow direct comparison with PP dynamics. Identifying dominant frequencies and time-varying cyclical behaviour is particularly important for evaluating whether Auto exhibits a more regular cycle structure and provides an independent benchmark for validating regime durations estimated later.

Figure 8: Smoothed spectrum – Auto loss ratio

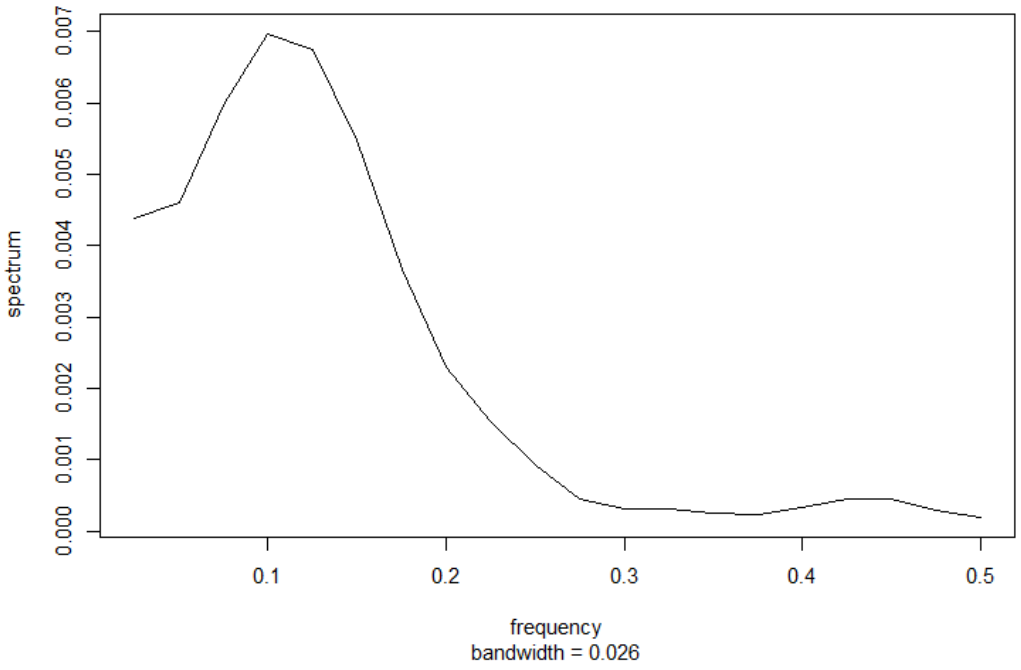
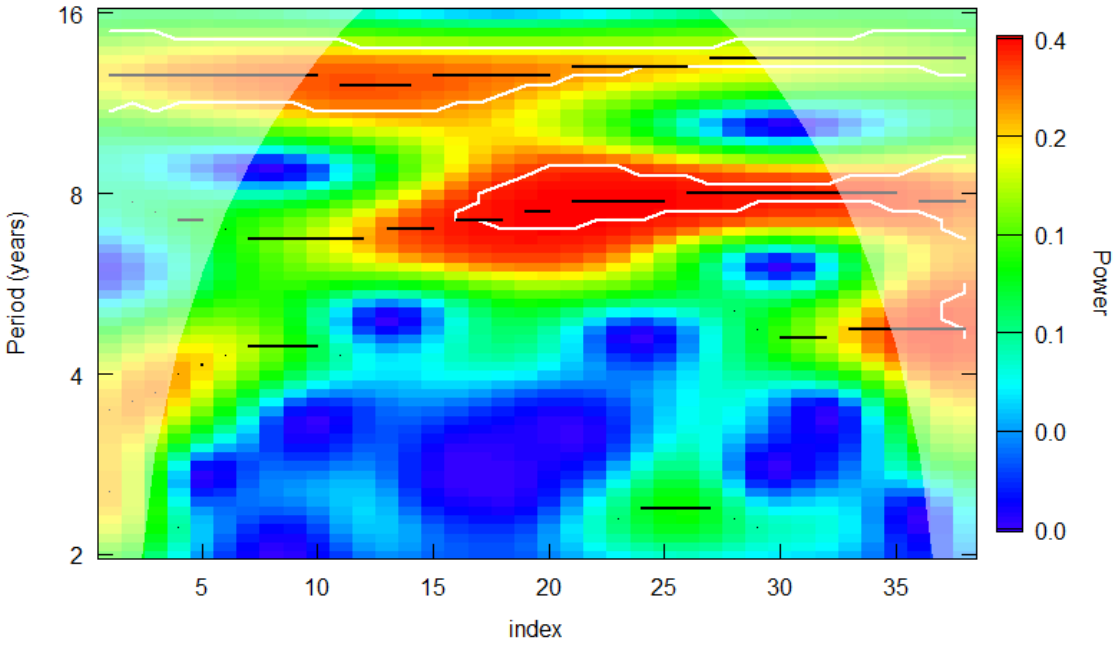


Figure 9: Wavelet power spectrum – Auto loss ratio



The spectral peak at frequency 0.1, corresponding to a 10-year period, is analytically distinct from PP. Unlike the diffuse intermittent power in PP, this sharp, well-defined peak signals a genuine recurring cycle rather than a transient artefact (Priestley, 1981; Venezian, 2006). Wavelet analysis confirms sustained, statistically significant power in the 7-to-10-year band throughout most of the observation window, consistent with three to four full cycles across the sample (Torrence and Compo, 1998). This independently derived 10-year cycle provides a critical external benchmark: if HMM transition probabilities in Section 4.6 align with this period, it constitutes strong cross-method validation that the model captures genuine underlying dynamics rather than circularly reproducing known frequencies (Boyer and Owadally, 2015).

4.6 Hidden Markov Models (HMM)

4.6.1 Personal Property

The HMM is applied to the PP loss ratio levels, for which prior diagnostics indicated persistent low-frequency variation. The objective is to identify economically meaningful market regimes and estimate their statistical characteristics. Transition probabilities and implied regime durations quantify the persistence of hard and soft market conditions.

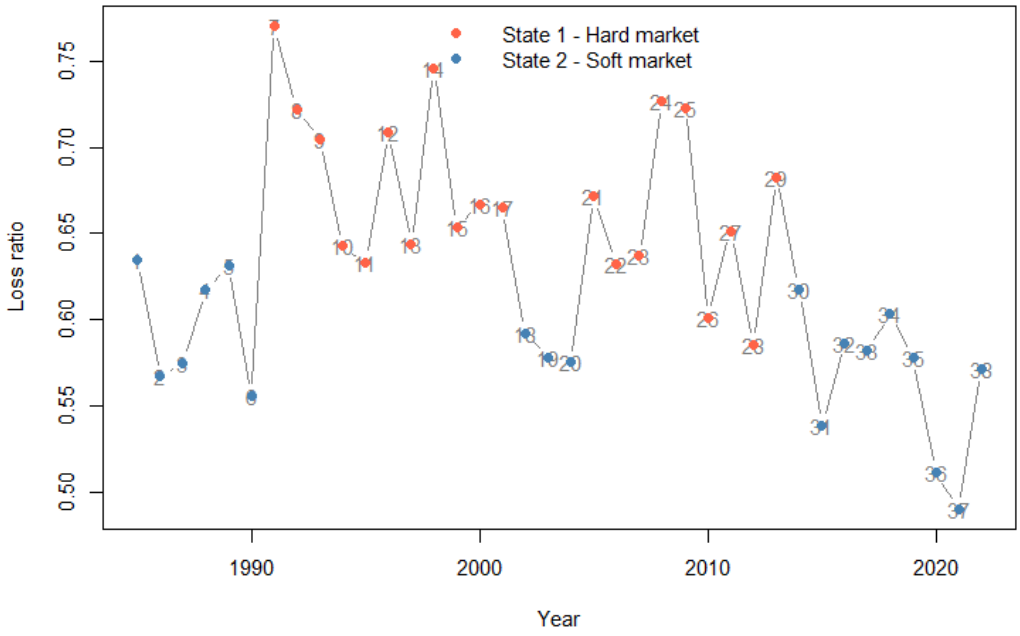
Table 5: HMM parameter estimates - PP

STATE	MEAN LR	SD LR	P(STAY)	P(SWITCH)
S1	0.668	0.052	0.90	0.10
S2	0.580	0.040	0.87	0.13

The two-state Gaussian HMM estimated directly on PP levels identifies two economically meaningful regimes. State 1 is a hard market (mean LR = 0.668, SD = 0.052) characterised by elevated but relatively stable underwriting losses. State 2 is a soft market (mean LR = 0.580, SD = 0.040) with stronger profitability and slightly lower variance.

The transition matrix shows strong persistence in both states: P(stay in S1) = 0.90 and P(stay in S2) = 0.87. This degree of persistence is consistent with the long-run asymmetry of the Canadian property market, where extended hard-market periods, particularly prominent from the late 1980s through the early 2000s, were associated with adverse weather experience, reserve pressures, and post-9/11 reinsurance tightening (Harrington, 2004).

Figure 10: Hidden Markov state classification of PP loss ratio over time



The decoded state path in Figure 10 aligns closely with major historical events, including the Calgary hailstorm (1991), the eastern Canada ice storm (1998), and post-9/11 reinsurance hardening (2001–2003). This correspondence supports the interpretation that detected regimes reflect genuine market conditions rather than statistical artefacts, addressing concerns raised by Boyer and Owadally (2015).

The residual based HMM fitted to AR(1) residuals as a robustness check yielded a degenerate solution with P(stay) = 0.002 for State 2, confirming that after removing AR(1) mean dynamics there is insufficient regime structure in the residuals and that the levels specification is the correct and only valid approach for PP (Harrington and Yu, 2003; Hamilton, 1989).

A three-state extension was estimated for completeness. The model identified a transitional State 2 (P(stay) = 0.544) between the hard and soft states, with States 1 and 3 unable to persist independently. While this transitional state is qualitatively plausible, representing years of gradual market adjustment, BIC favours the two-state specification (BIC = -89.63 versus -73.62 for the three-state), as the additional six parameters incur a penalty of 21.8 log likelihood units with n = 38 (Hyndman and Athanasopoulos, 2021). The two-state model is therefore the primary specification for PP.

4.6.2 Auto

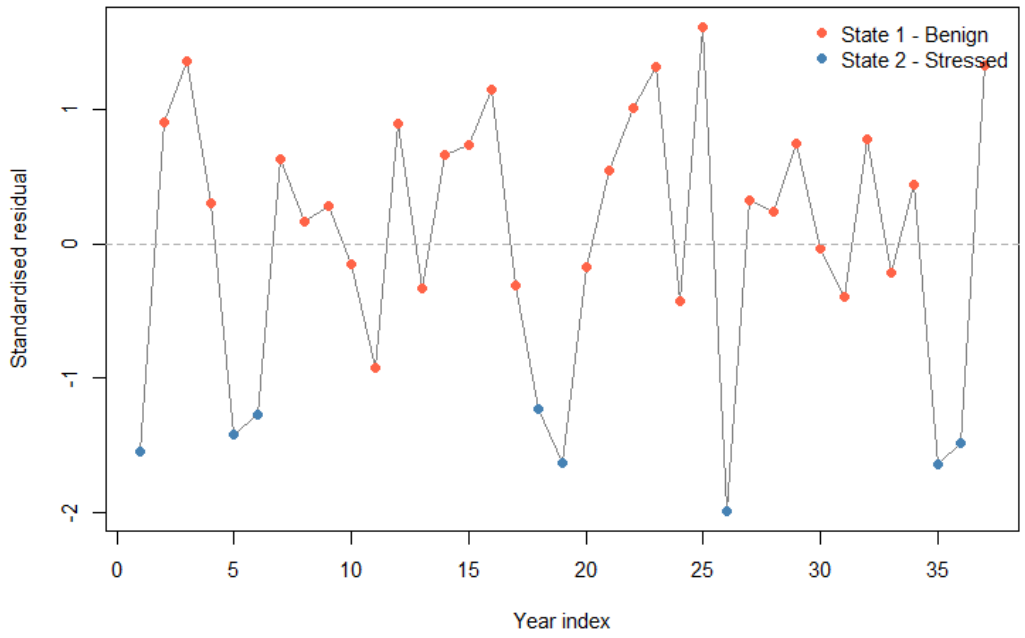
For the Auto series, the HMM is estimated using standardised residuals derived from the ARIMA–GARCH framework. This approach allows regime dynamics to be identified

after accounting for differencing and volatility persistence. The resulting regime structure provides insight into the frequency and duration of stressed versus benign operating conditions.

Table 6: HMM parameter estimates – Auto

STATE	MEAN RESIDUAL	SD RESIDUAL	P(STAY)	P(SWITCH)
S1	0.428	0.637	0.854	0.146
S2	-1.520	0.235	0.360	0.640

Figure 11: Auto GARCH residuals – HMM regime classification



For Auto, the HMM is applied to ARIMA(0,1,1)-GARCH(1,1) standardised residuals so that regimes capture behaviour beyond the fitted mean and volatility structure, following the approach recommended by Hamilton (1989) for integrated series. State 1 is a dominant benign regime (mean residual = 0.428, SD = 0.637, P(stay) = 0.854), representing typical years in which actual experience does not deviate systematically from the ARIMA-GARCH path. State 2 is a stressed regime (mean residual = −1.520, SD =

0.235, P(stay) = 0.360) corresponding to years with systematically worse than expected outcomes.

The decoded state path in Figure 11 aligns with identifiable Canadian Auto disruptions, including the Ontario accident benefit reforms (1994, 2010–2011) and pandemic-era claims volatility (2020–2022). The implied cycle length is also consistent with the ~10-year spectral peak identified earlier, providing cross-method validation of a decadal underwriting cycle (Torrence and Compo, 1998; Venezian, 2006).

The three-state Auto model failed to produce a valid solution, exhibiting degenerate transition probabilities indicative of EM instability. With only 38 annual observations and 12 parameters, this reflects over-parameterisation, and the negligible log-likelihood gain is offset by an essentially identical BIC (−82.59 versus −81.72), providing no evidence for additional regime complexity.

4.6.3 Regime Conditioned Forecasts and Practical Implications

The final stage of the analysis evaluates the practical forecasting implications of the estimated regime-switching models. This section also compares inferred regime durations with frequency-domain results to assess cross-method consistency.

Table 7: Expected regime durations and stationary distributions

SERIES	STATE	P(STAY)	E(DURATION)	LONG-RUN SHARE	INTERPRETATION
PP	S1 (Hard)	0.900	10.0 years	56.5%	Long hard market
	S2 (Soft)	0.870	7.7 years	43.5%	Extended soft market
AUTO	S1 (Benign)	0.854	6.8 years	81.4%	Dominant low risk
	S2 (Stressed)	0.360	1.6 years	18.6%	Short stress episodes

Table 7 summarises expected regime durations and long-run stationary distributions for both lines. The hard market expected duration of 10.0 years for PP independently corroborates the Auto spectral peak at frequency 0.1 (period = 10 years), providing convergent evidence across two separate analytical frameworks for a decadal Canadian P&C underwriting cycle. This cross-method validation directly meets the evidential bar

set by Boyer and Owadally (2015): the identified regimes are not statistical artefacts but genuine, independently detectable economic phenomena.

Figure 12: Regime-conditioned forecasts 2023–2025 - PP

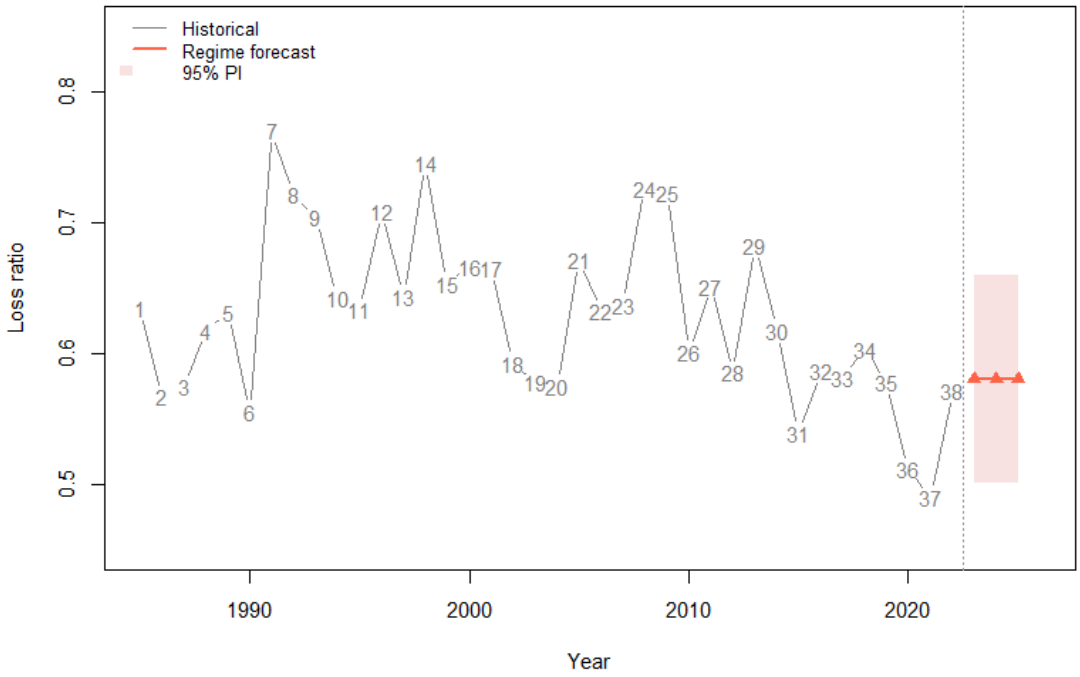
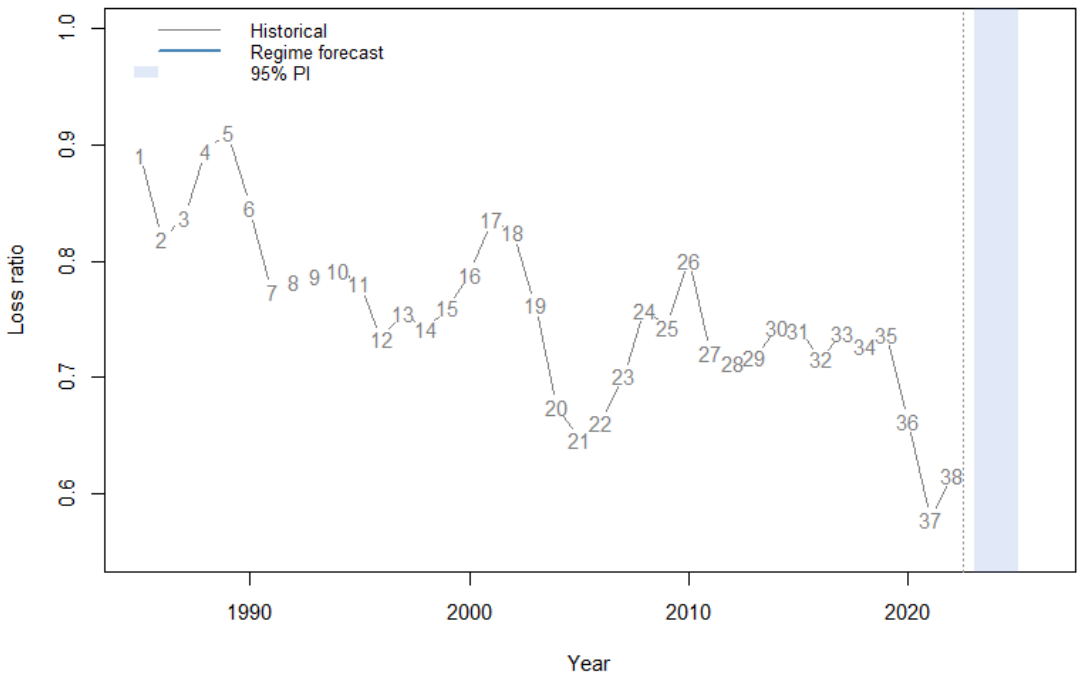


Figure 13: Regime-conditioned forecasts 2023–2025 – Auto



Figures 12 and 13 present regime-conditioned forecasts for PP and Auto for 2023–2025. Conditional on PP being in the soft regime at the end of 2022, the one-step-ahead forecast combines the two-state means weighted by transition probabilities, yielding a forecast loss ratio of 0.592. This is materially below the unconditional ARIMA forecast near the full-sample mean of 0.628, illustrating how regime information alters pricing signals. Although the probability of remaining in the soft state is high in the short run (87%), it declines to approximately 76% by year three, providing a transparent, model-based quantification of regime transition risk relevant for multi-year pricing and capital planning.

For Auto, Figure 13 shows a markedly different forecasting structure. Conditional on the benign regime prevailing at the end of the sample, forecast loss ratios remain close to the benign-state mean, with little deterministic trend over the forecast horizon. Risk in Auto arises not from cyclical mean reversion, but from discrete regime shifts: the estimated 14.6% annual probability of transitioning from benign to stressed manifests as forecast uncertainty rather than movement in the conditional mean. This transition probability provides a direct, internally consistent input to one-year risk assessments, such as Value-at-Risk calculations, without reliance on subjective expert judgement.

A formal out-of-sample evaluation using a rolling, expanding window was attempted but produced unreliable results due to the small training samples in early windows, a known limitation when applying HMMs with n = 38 annual observations. With quarterly data providing approximately 152 observations, stable rolling re-estimation and formal comparison via Diebold and Mariano (1995) tests for equal predictive accuracy would become feasible (Hyndman and Athanasopoulos, 2021). The second null hypothesis, therefore, remains formally untested and constitutes the primary direction for future work.

5. CONCLUSION

This project set out to answer whether Canadian P&C underwriting cycles are better characterised by persistent latent regimes than by stable linear dynamics, and whether HMM provides genuine forecasting improvements over ARIMA and GARCH benchmarks. The evidence strongly supports the first claim and provides partial grounds for the second.

The two-state HMM on PP levels recovers a hard market regime (mean LR = 0.668, expected duration 10.0 years) and a soft market regime (mean LR = 0.580, expected duration 7.7 years). The market spends 56.5% of its time in the harder state. For Auto, the HMM on ARIMA-GARCH residuals identifies a dominant benign state lasting 6.8 years, punctuated by brief stress episodes of 1.6 years. Crucially, the hard market's 10.0-year duration independently corroborates the decadal spectral cycle found for Auto. This provides convergent evidence from two separate analytical frameworks, transition matrix estimation and frequency domain analysis, for a decadal underwriting cycle in Canadian P&C insurance (Torrence and Compo, 1998; Hamilton, 1989). This cross-method validation directly addresses the concern of Boyer and Owadally (2015) that detected cycles may be statistical artefacts.

The primary limitation is sample size. With n = 38 annual observations, the three-state Auto model failed entirely due to over-parameterisation; rolling-window out-of-sample evaluation produced unreliable results; and time-varying transition probability extensions incorporating macroeconomic covariates such as interest rates and GDP growth (Grace and Hotchkiss, 1995) could not be estimated with confidence. The second null hypothesis, that HMM regime structure offers no out-of-sample forecast improvement, therefore remains formally untested. Quarterly data would provide approximately 152 observations, enabling three-state specifications, stable rolling evaluation, and formal Diebold and Mariano (1995) tests for equal predictive accuracy (Hyndman and Athanasopoulos, 2021). Extending the analysis to commercial property and liability lines would test whether the decadal cycle is a broad Canadian P&C phenomenon or specific to personal lines (Fung et al., 1998).

The underwriting cycle is not a smooth, predictable wave. It is a sequence of persistent market states, hard and soft conditions that can last a decade, governed by Markov transition dynamics that are quantifiable, historically coherent, and practically actionable. Hidden Markov Models are the appropriate tool for identifying, characterising, and projecting that structure, and this project demonstrates their value for the Canadian P&C underwriting cycle across two lines of business and two independent analytical frameworks.

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APPENDIX A – DATA SOURCE

The dataset used in this study is available in Excel format at the following link.

Canadian P&C Loss Ratio Dataset (1985–2022):

https://cityuni-my.sharepoint.com/:x:/g/personal/tisha_mendpara_bayes_city_ac_uk/IQAU2AQiobuITrFdwl0--5V6AVPlAo2tPgcg871hLNo1Vl0?e=XcIrQY

The dataset includes annual loss ratios, net incurred claims, and net written premiums extracted from multiple editions of the Insurance Bureau of Canada (IBC) Facts Books. The original source documents are accessible via the links below.

https://drive.google.com/file/d/11C_1DoKsc8jbP78WfEk03appCfJMI3Rt/view?usp=sharing

https://drive.google.com/file/d/1szD6W0as_eOu3zIihEaP4yOC1hxUjhT9/view?usp=sharing

https://drive.google.com/file/d/1fuv6u5wT_dVVm8Ckmm-DpFLBqT-FyNdj/view?usp=sharing

https://drive.google.com/file/d/17F24APnVOtHstWceR9Kyf-ZK_t5ERei9/view?usp=sharing

APPENDIX B – R CODE

The following R script was used to perform the statistical analysis and modelling of the dataset, including the calculation of key metrics and generation of output tables and figures.

R Script